

# Restaurant Spending Analysis

Plotly Express tips dataset | Python | Quantitative Analytics

## 1. Objective

Examine how restaurant bill size varies by day, smoking status and meal period, and test whether party si

## 2. Dataset and methods

The project uses the Plotly Express tips dataset containing 244 restaurant bills. Variables include total

## 3. Statistical workflow

The analysis calculates descriptive statistics by day, smoking status and meal period; applies a Welch in

## 4. Assumption checks

Normality is assessed using Shapiro-Wilk tests, variance equality using Levene test, and regression resid

## 5. Business interpretation

Party size is a natural operational driver of bill value and can inform table allocation, staffing and ca

## 6. Reproducibility

The repository contains analysis.py, the tips.csv dataset, supporting visualisations and this report.